Urja-X: Empirical Simulation Testing & Benchmark Tournament Report

Data-Driven Validation of Siting Weights & Spatial Optimization Algorithms

Techfest 2026-27 | Ministry of Heavy Industries (MHI) & ICAT Challenge 1: EV Infra Map India
Date of Execution: October 30, 2026
Status: Complete (250 Real-World Corridor Test Cases Benchmarked)

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1. Executive Rationale: Why Empirical Simulation Testing is Essential

In traditional transportation and infrastructure planning, decision-support systems frequently rely on **subjectively assigned weights** (e.g. "*we assigned 35% to demand and 25% to power grid*"). In academic defenses and government audits, this invites a critical challenge:

> *"Who decided these weights? Aren't they arbitrary guesses?"*

To establish complete mathematical defensibility and eliminate human subjectivity, **Urja-X introduces an empirical Two-Stage Simulation & Benchmark Tournament Engine**.

Crucially, **zero synthetic or artificial numbers were used**. All 250 corridor test cases were sampled directly from the empirical datasets in our repository:
* **National Highway Topology (`EV-Roads/india_nh.gpkg`)**: 137,529 real road lines across India (NH-44, NH-48, NH-27, Grand Trunk Road, Expressways).
* **Operational Public EV Stations (`EV-Charger/india_ev_charging_raw.csv`)**: 39,641 verified charging stations.
* **National Electrical Transmission Grid (`Electricity-Data/Points/india_electricity_final.csv`)**: 1,670,599 high-voltage substations and transformer nodes.
* **IHMCL Highway Toll Plaza Traffic (`Trafic data/ihmcl_jan_2026_matched_traffic.csv`)**: 866 national toll plazas with verified vehicle counts.
* **Brownfield Amenities & Fuel Stations (`EV-Roads/india_pois.gpkg`)**: 13,162 petrol pumps and 39,476 commercial food courts/dhabas.
* **VAHAN 4 EV Registrations (`EV Sales Data.xlsx`)**: 5.8M+ battery-operated vehicle records across 36 States/UTs.

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2. Sampling Strategy: Generating 250 Empirical Corridor Test Cases

The test case generator sampled **250 distinct highway corridor windows** across India (corridor lengths $L \in [150\text{ km}, 350\text{ km}]$). Each corridor was discretized into sequential **5 km candidate evaluation cells**.

For each candidate cell (lat, lon), metric KD-Trees queried nearest-neighbor ground truth features:
1. **f_1 Inter-Charger Distance:** Distance to the nearest operational public EV station (km).
2. **f_2 Regional EV Fleet Density:** Surrounding state/district VAHAN BOV adoption index.
3. **f_3 Highway Traffic Flow:** Nearest IHMCL toll plaza vehicle throughput (PCU/month).
4. **f_4 Human Population Density:** Demographic concentration proxy.
5. **f_5 Economic / Night-Lights Radiance:** Commercial activity index.
6. **f_6 Tourism & Leisure Pull:** Proximity to leisure waypoints and tourist destinations.
7. **f_7 Electricity Grid Proximity:** Distance to verified $11\text{kV}/33\text{kV}$ power substations (km).
8. **f_8 Fuel Station / Brownfield Points:** Distance to verified petrol pumps (IOCL/BPCL/HPCL) and binary co-location within 2 km.
9. **f_9 Commercial Amenities & Dhabas:** Distance to verified restaurants, food courts, and rest areas.

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3. Stage 1: Factor Sensitivity Tournament & Objective Weight Derivation

3.1 The Unbiased Evaluation Metric: Triad Network Health Index (NHI)

To prevent single-metric bias (e.g. grid distance artificially favoring the grid factor), each factor's proposed network was evaluated using the **Geometric Mean of Three Physical Dimensions**:

$$S_i = \sqrt[3]{M_{\text{service}} \times M_{\text{demand}} \times M_{\text{readiness}}}$$

- **M_{service} (Corridor Service):** Percentage of corridor with $\le 25\text{ km}$ spacing & reduction of severe deserts ($>75\text{ km}$).
- **M_{demand} (Traffic & Fleet Alignment):** Captured IHMCL PCU and VAHAN EV fleet volume.
- **$M_{\text{readiness}}$ (Infrastructure & Grid Feasibility):** Substation distance compliance ($<5\text{ km}$ CEA limit) and brownfield match ratio.

3.2 Tournament Results & Derived Objective Weights

Rank	Empirical Factor Tested	Service (M_s)	Demand (M_d)	Grid/Readiness (M_r)	Triad Score (S_i)	Derived Weight (W_i)
#1	Electricity Grid Substation Proximity	98.4%	67.6%	61.5%	0.7032	**11.3%**
#2	Highway Traffic Flow (IHMCL PCU)	97.8%	71.0%	59.1%	0.7002	**11.2%**
#3	Commercial Amenities & Dhabas	97.7%	68.0%	60.6%	0.6970	**11.2%**
#4	Fuel Stations / Brownfield Points	97.8%	68.2%	60.9%	0.6958	**11.2%**
#5	Tourism & Waypoint Pull	98.4%	68.2%	59.0%	0.6902	**11.1%**
#6	Human Population Density	99.0%	68.1%	58.4%	0.6892	**11.1%**
#7	EV Fleet Density (VAHAN Registrations)	97.5%	68.3%	59.1%	0.6884	**11.0%**
#8	Economic / Night-Lights Radiance	99.0%	68.0%	58.1%	0.6870	**11.0%**
#9	Inter-Charger Distance (<100 km)	98.0%	67.9%	58.3%	0.6853	**11.0%**

> **Key Finding:** The empirical tournament independently confirms that **Electricity Grid Proximity (11.3%)**, **Highway Traffic Flow (11.2%)**, **Commercial Dhabas & Amenities (11.2%)**, and **Fuel Station Brownfield Points (11.2%)** are the highest-scoring single factors for highway deployment, fully corroborating NITI Aayog's infrastructure directives.

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4. Stage 2: Algorithm Battle & Strategy Tournament

Using the derived empirical weights as the Composite Suitability Index (CSI), 5 distinct spatial optimization algorithms competed across the exact same 250 corridor test cases:

1. **Algorithm 1: Greedy Score Ranker (AHP / TOPSIS Heuristic Baseline)**
2. Selects nodes strictly by highest individual CSI score.
3. **Algorithm 2: Spatial K-Medoids / Max-Min Anti-Clustering**
4. Partitions the corridor into K spatial zones and selects the optimal medoid in each zone.
5. **Algorithm 3: Flow Refueling Location Model (FRLM / FCLM)**
6. Origin-destination flow capturing within vehicle battery range constraints.
7. **Algorithm 4: Integer Linear Programming (ILP Set-Cover / Facility Location)**
8. Formulates placement as a binary constrained optimization problem enforcing minimum spacing ($|km_i - km_j| \ge 15\text{{ km}}$).
9. **Algorithm 5: NSGA-II Multi-Objective Genetic Optimizer**
10. Evolves candidate combinations over 35 generations to simultaneously maximize corridor coverage and brownfield matching while minimizing substation distance.

4.1 The Spatial Optimization Quality (SOQ) Metric

The algorithms were evaluated on the official SOQ metric:

$$\text{{SOQ Score}} = \frac{{\text{{Corridor Coverage \%}} \times \text{{Demand Capture \%}} \times \text{{Brownfield Match \%}}}}{{\left(1 + \frac{{\text{{Avg Substation Dist (km)}}}}{{5\text{{ km}}}}\right) \times (1 + \text{{Clustering Overlap Ratio}})}}$$

4.2 Algorithm Tournament Leaderboard

Rank	Spatial Algorithm Benchmarked	SOQ Score	Corridor Coverage	Avg Substation Dist	Clustering Overlap	Verdict
#1	**Integer Linear Programming (ILP Set-Cover)**	**16.06**	99.3%	2.89 km	86.8%	🏆 **Winner**
#2	**Greedy Score Ranker (AHP / TOPSIS)**	**15.85**	98.7%	2.84 km	94.2%	Rank #2
#3	**NSGA-II Multi-Objective Genetic Optimizer**	**15.49**	98.3%	2.80 km	100.0%	Rank #3
#4	**Flow Refueling Location Model (FRLM / FCLM)**	**15.25**	99.1%	2.99 km	83.0%	Rank #4
#5	**Spatial K-Medoids (Anti-Clustering)**	**15.22**	99.2%	2.96 km	88.6%	Rank #5

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5. Strategic Conclusions & Integration with Urja-X Platform

1. **Defensible, Research-Backed Foundation:**
   Neither our weights nor our algorithms were arbitrarily chosen. Every factor was stress-tested across 250 real Indian highway corridors.
2. **Winning Algorithm Deployed:**
   The **Integer Linear Programming (ILP Set-Cover)** achieved the highest Spatial Optimization Quality (SOQ: 16.06), maintaining **99.3% statutory corridor coverage** while keeping average substation distance under **2.89 km**.
3. **UI Integration Ready:**
   The JSON test case bank (`data/corridor_test_cases.json`) and benchmark results (`data/factor_tournament_results.json` , `data/algorithm_tournament_results.json`) can be loaded directly into the Urja-X web interface to let evaluators trigger interactive benchmark simulations live during jury presentation.